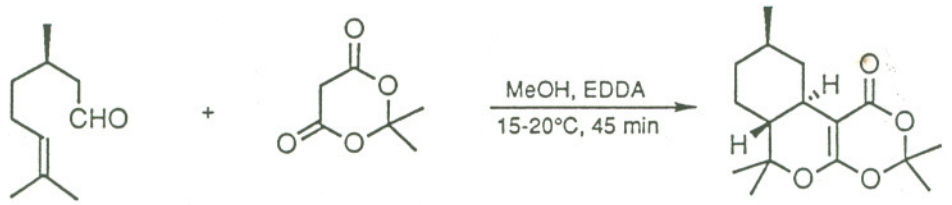
ORGANIC CUMULATIVE EXAMINATION
November 10, 1990

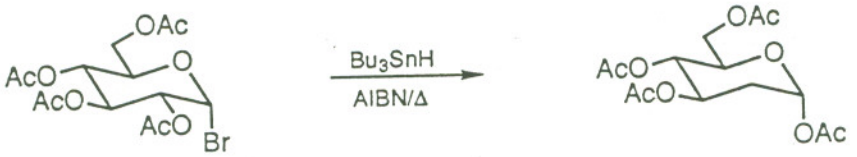
Propose a reasonable step by step mechanism for each of the following reactions or reaction sequences. Show structures of all significant mechanistic intermediates, and use arrows to portray electron flow in bond-forming and bond-breaking steps. Briefly explain stereoselectivity in terms of pertinent structural and/or mechanistic principles.

1.



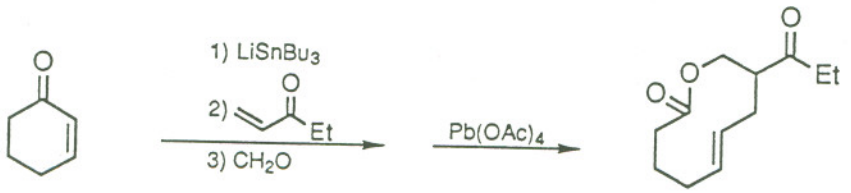
(EDDA = ethylenediammonium diacetate)

2.



(AIBN = 2,2′-azobisisobutyronitrile)

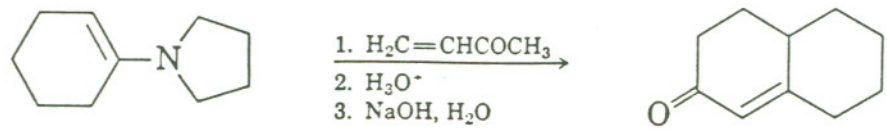
3.



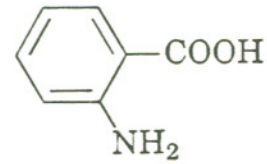
(Steps 1, 2 and 3 are carried out sequentially in the same flask without isolation of intermediates. The crude product from that sequence is isolated and then treated separately with $Pb(OAc)_4$ in benzene.)

Reference: Organic Syntheses, 69, 31, 66, 188 (1990).

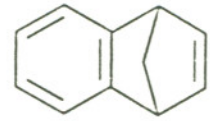
23.50 The Stork enamine reaction and the intramolecular aldol reaction can be carried out in sequence to allow the synthesis of cyclohexenone rings. For example, reaction of the pyrrolidine enamine of cyclohexanone with 3-buten-2-one, followed by enamine hydrolysis and base treatment yields the product indicated. Show the mechanisms of the different steps.



26.29 Reaction of anthranilic acid (*o*-aminobenzoic acid) with $NaNO_2$ and H_2SO_4 yields a diazonium salt that can be treated with base to yield a neutral diazonium carboxylate.



Anthranilic acid

(a) What is the structure of the neutral diazonium carboxylate?

(b) Heating the diazonium carboxylate results in the formation of CO_2, N_2, and a high-energy intermediate that reacts with 1,3-cyclopentadiene to yield the following organic product: